CHAPTER 5

WORK, ENERGY AND POWER

CHAPTER ESSENCE AT A GLANCE:
Work is just energy transferring from one wallet to another! When work is done against gravity or a spring, it gets saved as Potential Energy. Total mechanical energy (KE + PE) remains strictly constant in any closed conservative system!

4 CORE PRINCIPLES TO UNDERSTAND FIRST

• **Work Definition:** W = F · d = F d cos θ. Work is zero if force is perpendicular to displacement (θ = 90°, cos 90° = 0), like gravity on a satellite in circular orbit.

• **Conservative Forces:** Work done depends only on initial and final positions, and is strictly ZERO in any closed loop (Gravity, Spring force). Non-conservative forces (Friction) dissipate energy as heat (work in loop ≠ 0).

• **Work-Energy Theorem:** Net work done by all forces equals the change in kinetic energy: W net = ΔK = ½ m v^2 - ½ m u^2 (proved via W = ∫ F dx).

• **Mechanical Energy Conservation:** In a conservative system, Total Energy E = K + U = constant. In free fall, E A = E B = E C = mgh. Spring potential energy U(x) = ½ k x^2.

ESSENTIAL FORMULAS & GOVERNING EQUATIONS

$$W = \int \mathbf{F} \cdot d\mathbf{r} = \Delta K = \tfrac{1}{2}mv^2 - \tfrac{1}{2}mu^2$$

Work-Energy Theorem for Variable Force

$$U(x) = \tfrac{1}{2}kx^2, \quad F = -kx$$

Elastic Potential Energy of Stretched Spring

$$P = \frac{dW}{dt} = \mathbf{F} \cdot \mathbf{v}$$

Instantaneous Power as Scalar Product

$$v_1' = \left(\frac{m_1 - m_2}{m_1 + m_2}\right)u_1 + \left(\frac{2m_2}{m_1 + m_2}\right)u_2$$

1D Elastic Collision Final Velocity

CRUCIAL EXAM PRECAUTIONS & VALUATION RUBRIC SECRETS

• **Alert:** No work is done if force is perpendicular to displacement (θ = 90°, cos 90° = 0)! E.g. Earth revolving around the Sun, or carrying a bag horizontally.

• **Alert:** 1 Horsepower (HP) = 746 Watts. Remember this for 1-mark objective questions!

HIGH-YIELD CENTUM QUESTIONS & STEP-BY-STEP EXPLANATIONS

1. Distinguish between conservative forces and non-conservative forces with one example of each. [Score: 2 Marks]

Answer & Key Valuation Steps:

• **[Step 1: Conservative Forces]** Work done is independent of the path taken and depends only on initial and final positions; work done in a closed round loop is zero (oint F . dr = 0). Example: Gravitational force, electrostatic force. [1 Mark]

• **[Step 2: Non-Conservative Forces]** Work done depends on the actual path taken; energy is dissipated as heat/sound and oint F . dr ≠ 0. Example: Frictional force, viscous drag. [1 Mark]

• **Key Exam Result:** Conservative (path-independent, gravity) vs Non-conservative (path-dependent, friction)

2. Define power. State its SI unit and commercial unit (kWh), and relate 1 kWh to Joules. [Score: 2 Marks]

Answer & Key Valuation Steps:

• **[Step 1: Power Definition & SI Unit]** Power is the rate of doing work or rate of energy transfer: P = W divided by t. SI unit: Watt (W = J s^{-1}). [1 Mark]

• **[Step 2: Commercial Unit & Joule Relation]** Commercial unit: kilowatt-hour (kWh). 1 kWh = 1 kW x 1 h = 1000 W x 3600 s = 3.6 × 10^6 Joules. [1 Mark]

• **Key Exam Result:** P = W divided by t (Watt); 1 kWh = 3.6 × 10^6 J

3. State and prove the Work-Energy Theorem for a constant force in one dimension. [Score: 3 Marks]

Answer & Key Valuation Steps:

• **[Step 1: Statement]** The work done by the net force acting on a particle equals the change in its kinetic energy: W net = ΔK = K f - K i. [1 Mark]

• **[Step 2: Proof for Constant Force]** From kinematic equation $v^2 - u^2 = 2 a s$, multiplying both sides by m divided by 2: ½ m ($v^2 - u^2$) = m a s. [1 Mark]

• **[Step 3: Equating to Work]** Since F = m a: ½ m v^2 - ½ m u^2 = F s. Since W = F s: K f - K i = W => W = ΔK. [1 Mark]

• **Key Exam Result:** W net = ΔK = ½ m v^2 - ½ m u^2

4. Prove the Work-Energy Theorem for a variable force using calculus. [Score: 3 Marks]

Answer & Key Valuation Steps:

• **[Step 1: Kinetic Energy Rate of Change]** K = ½ m v^2. Differentiating with respect to t: dK divided by dt = ½ m (2 v dv divided by dt) = m v (dv divided by dt). [1 Mark]

• **[Step 2: Relating to Force and Displacement]** Since dv divided by dt = a and v = dx divided by dt: dK divided by dt = m a (dx divided by dt) = F (dx divided by dt) => dK = F dx. [1 Mark]

• **[Step 3: Integration]** Integrating from initial position x i (speed u) to final position x f (speed v): $\int_{K\,i}^{K\,f} dK = \int_{x\,i}^{x\,f} F\,dx$ => K f - K i = W => ΔK = W. [1 Mark]

• **Key Exam Result:** W = ΔK (Proved for variable force)

5. Derive an expression for the potential energy stored in an elastic stretched spring of spring constant k: U = ½ k x^2. [Score: 3 Marks]

Answer & Key Valuation Steps:

• **[Step 1: Hooke's Law Restoring Force]** According to Hooke's law, spring restoring force is F s = -k x, where x is displacement from equilibrium. The external force required to stretch the spring without acceleration is F ext = +k x. [1 Mark]

• **[Step 2: Work Done during Extension]** Small work done for extension dx: dW = F ext dx = k x dx. [1 Mark]

• **[Step 3: Total Work & Potential Energy]** W = $\int_0^x$ k x dx = k [x^2 divided by 2] 0^x = $\frac{1}{2}$ k x^2. This work is stored as elastic potential energy: U(x) = $\frac{1}{2}$ k x^2. [1 Mark]

• **Key Exam Result:** Spring Potential Energy U = ½ k x^2

6. A pump on the ground floor of a building can pump water to fill a tank of volume 30 m^3 in 15 minutes. The tank is 40 m above the ground. If the efficiency of the pump is 30%, find the electric power consumed. (Take g = 9.8 m/s^2, density of water = 1000 kg/m^3) **[Score: 3 Marks]**

Answer & Key Valuation Steps:

• **[Step 1: Mass and Work Done]** Volume V = 30 m^3 ⇒ Mass m = V x ρ = 30 × 1000 = 30,000 kg. Height h = 40 m. Work output W = m g h = 30000 × 9.8 × 40 = 1.176 × 10^7 J. [1 Mark]

• **[Step 2: Output Power]** Time t = 15 min = 15 × 60 = 900 s. Output power P out = W divided by t = (1.176 × 10^7) divided by 900 = 13,066.7 W = 13.07 kW. [1 Mark]

• **[Step 3: Input Electric Power]** Efficiency η = P out divided by P in = 0.30 => P in = P out divided by 0.30 = 13066.7 divided by 0.30 = 43,555.6 W = 43.56 kW. [1 Mark]

• **Key Exam Result:** Consumed Power = 43.56 kW

7. Prove that the total mechanical energy of a body falling freely under gravity is conserved at all points of its motion. **[Score: 4 Marks]**

Answer & Key Valuation Steps:

• **[Step 1: Point A (Highest Point at height h)]** Velocity v A = 0. Kinetic energy K A = 0. Potential energy U A = mgh. Total mechanical energy E A = K A + U A = 0 + mgh = mgh ---(1). [1 Mark]

• **[Step 2: Point B (Fallen distance x)]** Height above ground = h - x. Velocity v B^2 = 0 + 2gx = 2gx. Kinetic energy K B = $\frac{1}{2}$ m v B^2 = $\frac{1}{2}$ m (2gx) = mgx. Potential energy U B = mg(h - x). Total energy E B = K B + U B = mgx + mgh - mgx = mgh ---(2). [1.5 Marks]

• **[Step 3: Point C (Ground Level)]** Height = 0 ⇒ U C = 0. Velocity v C^2 = 0 + 2gh = 2gh. Kinetic energy K C = $\frac{1}{2}$ m (2gh) = mgh. Total energy E C = K C + U C = mgh + 0 = mgh ---(3). Since E A = E B = E C = mgh, total mechanical energy is conserved. [1.5 Marks]

• **Key Exam Result:** E total = mgh = Constant everywhere